AYMANE AIT DADS

AI Research Intern - Embodied Robotics | Multimodal Perception · LLM Fine-Tuning · Computer Vision

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PROFESSIONAL SUMMARY

Top-10% Kaggle competitor fine-tuning a 30B Mamba-Transformer MoE for multi-step logical reasoning, directly demonstrating the model architecture and ablation methodology central to embodied AI research. Ranked 1st in class on an international AI-generated image detection benchmark and submitted to NTIRE 2026 @ CVPR, achieving 0.791 AUC with a CLIP ViT-L/14 multimodal backbone across 250K training images. Core competencies include LLM fine-tuning with LoRA/PEFT, computer vision with transformer architectures, simulation-compatible sensor data pipelines, and rigorous single-variable ablation studies. Available from Summer 2025 for internship in Singapore; holds no visa restrictions for Asia-Pacific placement.

CORE COMPETENCIES

Embodied AI Research | Multimodal Perception | Simulation & Robotics | LLM Fine-Tuning | Ablation Studies | Computer Vision | Sensor Validation | Algorithm Prototyping

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Engineered a **Random Forest + K-Means pipeline** on 100K+ fiber-optic network records, classifying signal quality across thousands of sensor nodes for the network optimization team.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, directly informing infrastructure maintenance decisions from multi-dimensional sensor data.
- Reduced manual network analysis time by **~60%** through automated feature engineering and systematic model evaluation across upload speed, latency, and jitter dimensions.
- Delivered stakeholder presentations translating **model outputs** into actionable maintenance recommendations, bridging algorithm research and real-world operational deployment.

PROJECTS

NVIDIA Nemotron Reasoning Challenge *Kaggle Competition · In Progress, June 2026 · Top 10% Public Leaderboard*

- Achieved **top 10% public leaderboard (0.80/1.0)** in a $106K+ prize-pool competition by fine-tuning a Nemotron-3-Nano-30B hybrid Mamba-Transformer MoE (3.5B active params via MoE routing) with LoRA (r=32, ~888M trainable parameters) on 7,828 chain-of-thought reasoning examples.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** via monkey-patching, sys.modules stubbing, and pure-PyTorch fallbacks, enabling training on 95GB VRAM hardware; conducted 20+ single-variable ablations confirming completion-only loss (+0.01 AUC) and that sequence packing catastrophically degrades out-of-distribution generalization (0.80 to 0.64).

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection - NTIRE 2026 @ CVPR *EURECOM ImSecu + NTIRE 2026 @ CVPR CodaBench · 2025 · Ranked 1st in Class*

- Ranked **1st in EURECOM class** on a private Kaggle leaderboard (0.791 AUC) by fine-tuning a CLIP ViT-L/14 multimodal backbone (428M params, pretrained on 400M image-text pairs) with a custom MLP head across 250K images spanning 25+ generative model types; also submitted independently to the NTIRE 2026 @ CVPR CodaBench international benchmark.
- Identified through **ablation study** that 1-epoch training (0.793 AUC) outperforms 4-epoch (0.767 AUC) due to out-of-distribution generalization; deployed a 3-model ensemble weighted by validation AUC with 10-view TTA (flips, crops, blur, resizes) for final submission.

PyTorch · OpenCLIP · Hugging Face Transformers · FP16 Mixed Precision

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM *Sep 2023 - Present*

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Image Security, Cloud Computing, Statistics

CPGE - Mathematics & Physics - Ibn Timiya *2021 - 2023*

SKILLS

LLM Fine-Tuning & Multimodal AI: LoRA / QLoRA | SFTTrainer (Unsloth) | PEFT | Mamba-Transformer MoE | vLLM | Chain-of-Thought | Ablation Studies

Computer Vision & Deep Learning: CLIP ViT-L/14 | CNN / DenseNet | PyTorch | TensorFlow | Scikit-learn | FP16 Mixed Precision | TTA Ensembling

Data & MLOps: Python | Pandas / NumPy | SQL | Git / Linux | CUDA Debugging | Triton | Power BI | Docker (basic)